In [1]:
```
import pandas as pd
```

In [2]:
```
movies=pd.read_csv(r"C:\Users\Aisharani Patro\OneDrive\Desktop\movie.csv")
movies
```

Out[2]:

	movieId	title	genres
0	1	Toy Story (1995)	Adventure|Animation|Children|Comedy|Fantasy
1	2	Jumanji (1995)	Adventure|Children|Fantasy
2	3	Grumpier Old Men (1995)	Comedy|Romance
3	4	Waiting to Exhale (1995)	Comedy|Drama|Romance
4	5	Father of the Bride Part II (1995)	Comedy
...	...	...	...
27273	131254	Kein Bund für's Leben (2007)	Comedy
27274	131256	Feuer, Eis & Dosenbier (2002)	Comedy
27275	131258	The Pirates (2014)	Adventure
27276	131260	Rentun Ruusu (2001)	(no genres listed)
27277	131262	Innocence (2014)	Adventure|Fantasy|Horror

27278 rows × 3 columns

In [3]:
```
movies.shape
```

Out[3]:
```
(27278, 3)
```

In [4]:
```
ratings=pd.read_csv(r"C:\Users\Aisharani Patro\OneDrive\Desktop\rating.csv")
ratings
```

Out[4]:

	userId	movieId	rating	timestamp
0	1	2	3.5	2005-04-02 23:53:47
1	1	29	3.5	2005-04-02 23:31:16
2	1	32	3.5	2005-04-02 23:33:39
3	1	47	3.5	2005-04-02 23:32:07
4	1	50	3.5	2005-04-02 23:29:40
...	...	...	...	...
20000258	138493	68954	4.5	2009-11-13 15:42:00
20000259	138493	69526	4.5	2009-12-03 18:31:48
20000260	138493	69644	3.0	2009-12-07 18:10:57
20000261	138493	70286	5.0	2009-11-13 15:42:24
20000262	138493	71619	2.5	2009-10-17 20:25:36

20000263 rows × 4 columns

In [5]:
```
ratings.shape
```

Out[5]:
```
(20000263, 4)
```

In [6]:
```
ratings.head(3)
```

Out[6]:

	userId	movieId	rating	timestamp
0	1	2	3.5	2005-04-02 23:53:47
1	1	29	3.5	2005-04-02 23:31:16
2	1	32	3.5	2005-04-02 23:33:39

In [7]:
```
ratings.columns
```

Out[7]:
```
Index(['userId', 'movieId', 'rating', 'timestamp'], dtype='object')
```

In [8]:
```
tag=pd.read_csv(r"C:\Users\Aisharani Patro\OneDrive\Desktop\tag.csv")
tag
```

Out[8]:

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	2009-04-24 18:19:40
1	65	208	dark hero	2013-05-10 01:41:18
2	65	353	dark hero	2013-05-10 01:41:19
3	65	521	noir thriller	2013-05-10 01:39:43
4	65	592	dark hero	2013-05-10 01:41:18
...	...	...	...	...
465559	138446	55999	dragged	2013-01-23 23:29:32
465560	138446	55999	Jason Bateman	2013-01-23 23:29:38
465561	138446	55999	quirky	2013-01-23 23:29:38
465562	138446	55999	sad	2013-01-23 23:29:32
465563	138472	923	rise to power	2007-11-02 21:12:47

465564 rows × 4 columns

In [9]:
```
tag.shape
```

Out[9]:
```
(465564, 4)
```

In [10]:
```
tag.head(4)
```

Out[10]:

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	2009-04-24 18:19:40
1	65	208	dark hero	2013-05-10 01:41:18
2	65	353	dark hero	2013-05-10 01:41:19
3	65	521	noir thriller	2013-05-10 01:39:43

In [11]:
```
row_0=tag.iloc[0]
type(row_0)
```

Out[11]:
```
pandas.core.series.Series
```

In [12]:
```
print(row_0)
```

```
userId                       18
movieId                    4141
tag                 Mark Waters
timestamp    2009-04-24 18:19:40
Name: 0, dtype: object
```

In [13]:
```
row_0.index
```

Out[13]: Index(['userId', 'movieId', 'tag', 'timestamp'], dtype='object')

In [14]:
```
print(row_0['userId'])
```

```
18
```

In [15]:
```
'rating' in row_0
```

Out[15]: False

In [16]:
```
'tag' in row_0
```

Out[16]: True

In [17]:
```
row_0.shape
```

Out[17]: (4,)

In [18]:
```
row_0.name
```

Out[18]: 0

In [19]:
```
row_0=row_0.rename('firstRow')
row_0.name
```

Out[19]: 'firstRow'

In [20]:
```
row_0
```

Out[20]:
```
userId                        18
movieId                     4141
tag                  Mark Waters
timestamp    2009-04-24 18:19:40
Name: firstRow, dtype: object
```

In [21]:
```
tag.head()
```

Out[21]:

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	2009-04-24 18:19:40
1	65	208	dark hero	2013-05-10 01:41:18
2	65	353	dark hero	2013-05-10 01:41:19
3	65	521	noir thriller	2013-05-10 01:39:43
4	65	592	dark hero	2013-05-10 01:41:18

In [22]:
```
tag.index
```

Out[22]: RangeIndex(start=0, stop=465564, step=1)

In [23]:
```
tag.columns
```

Out[23]: Index(['userId', 'movieId', 'tag', 'timestamp'], dtype='object')

In [24]:
```
tag.iloc[ [0,11,500] ]
```

Out[24]:

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	2009-04-24 18:19:40
11	65	1783	noir thriller	2013-05-10 01:39:43
500	342	55908	entirely dialogue	2012-01-31 18:41:16

In [25]:
```
ratings['rating'].describe()
```

Out[25]:
```
count    2.000026e+07
mean     3.525529e+00
std      1.051989e+00
min      5.000000e-01
25%      3.000000e+00
50%      3.500000e+00
75%      4.000000e+00
max      5.000000e+00
Name: rating, dtype: float64
```

In [26]:
```
ratings.describe()
```

Out[26]:

	userId	movieId	rating
count	2.000026e+07	2.000026e+07	2.000026e+07
mean	6.904587e+04	9.041567e+03	3.525529e+00
std	4.003863e+04	1.978948e+04	1.051989e+00
min	1.000000e+00	1.000000e+00	5.000000e-01
25%	3.439500e+04	9.020000e+02	3.000000e+00
50%	6.914100e+04	2.167000e+03	3.500000e+00
75%	1.036370e+05	4.770000e+03	4.000000e+00
max	1.384930e+05	1.312620e+05	5.000000e+00

In [27]:
```
ratings['rating'].mean()
```

Out[27]: np.float64(3.5255285642993797)

In [28]:
```
ratings['rating'].min()
```

Out[28]: 0.5

In [29]:
```
ratings['rating'].max()
```

Out[29]: 5.0

In [30]:
```
ratings['rating'].std()
```

Out[30]: 1.051988919275684

In [31]:
```
ratings['rating'].mode()
```

Out[31]:
```
0    4.0
Name: rating, dtype: float64
```

In [32]:
```
ratings[['userId','movieId','rating']].corr()
```

Out[32]:

	userId	movieId	rating
userId	1.000000	-0.000850	0.001175
movieId	-0.000850	1.000000	0.002606
rating	0.001175	0.002606	1.000000

In [33]:
```
filter1=ratings['rating']>10
print(filter1)
filter1.any()
```

```
0           False
1           False
2           False
3           False
4           False
            ...
20000258    False
20000259    False
20000260    False
20000261    False
20000262    False
Name: rating, Length: 20000263, dtype: bool
```

Out[33]: np.False_

In [34]:
```
filter2=ratings['rating']>0
filter2.all()
```

Out[34]: np.True_

In [35]:
```
movies.shape
```

Out[35]: (27278, 3)

In [36]:
```
movies.isnull().sum()
```

Out[36]:
```
movieId    0
title      0
genres     0
dtype: int64
```

In [37]:
```
ratings.isnull().any().any()
```

Out[37]:
```
np.False_
```

In [38]:
```
tag.shape
```

Out[38]:
```
(465564, 4)
```

In [39]:
```
tag.isnull().sum()
```

Out[39]:
```
userId        0
movieId       0
tag          16
timestamp     0
dtype: int64
```

In [40]:
```
tag=tag.dropna()
```

In [41]:
```
tag.isnull().sum()
```

Out[41]:
```
userId       0
movieId      0
tag          0
timestamp    0
dtype: int64
```

In [42]:
```
tag.isnull().any().any()
```

Out[42]:
```
np.False_
```

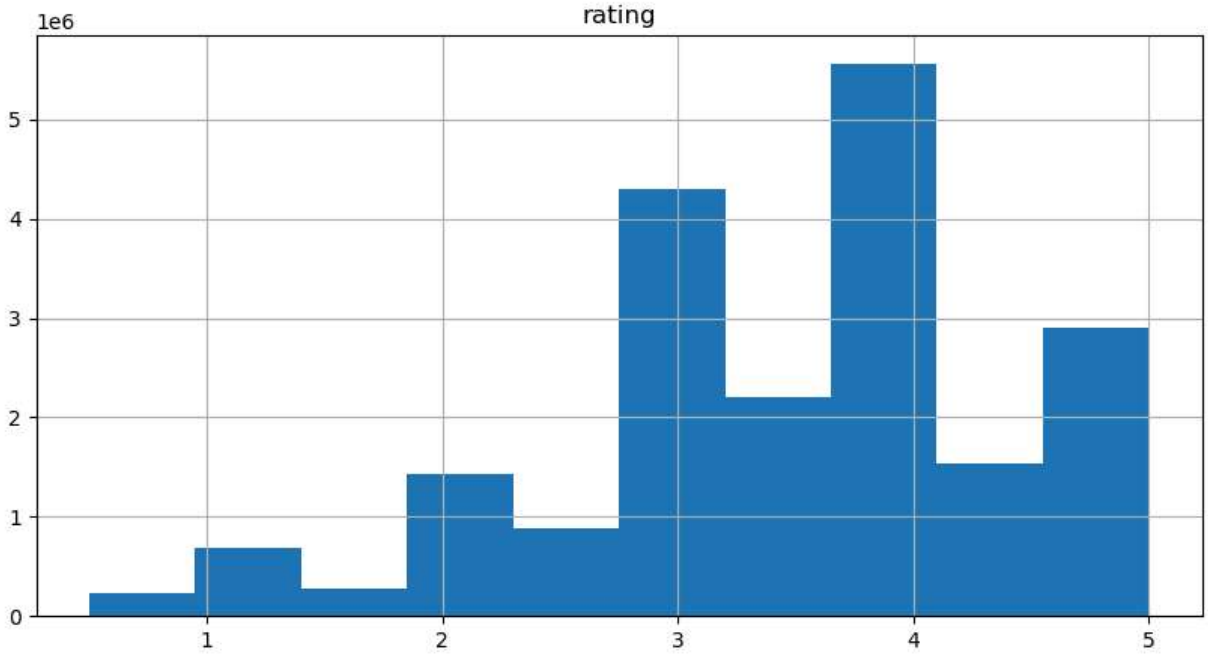
In [43]:
```
tag.shape
```

Out[43]:
```
(465548, 4)
```

In [44]:
```
%matplotlib inline
```

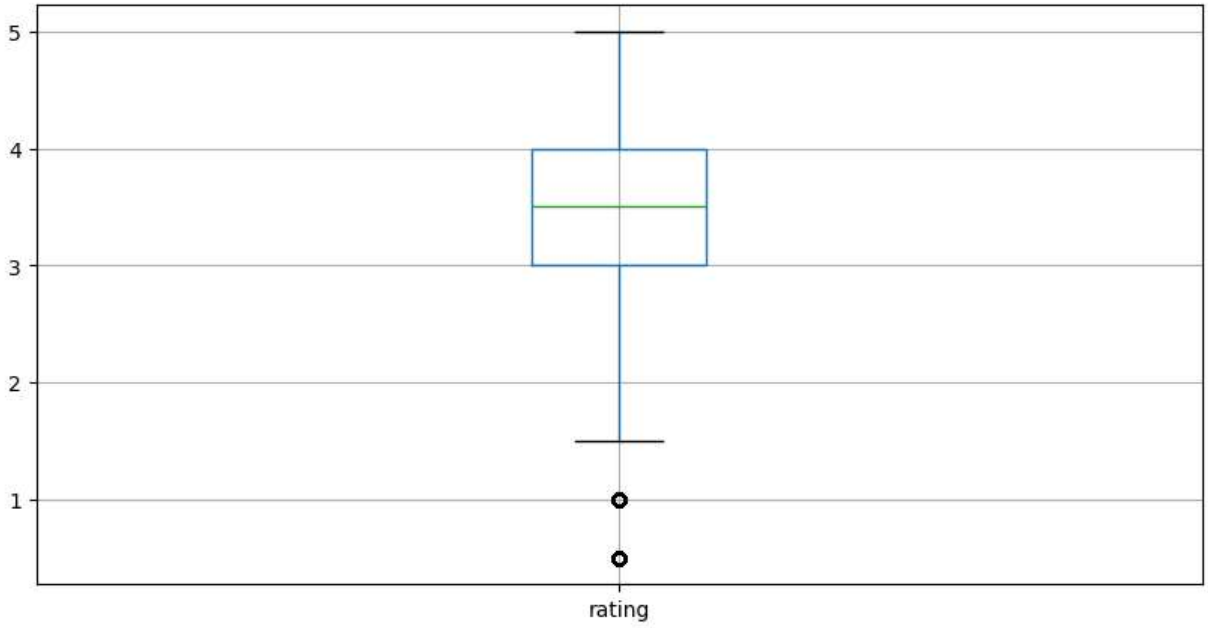
In [45]:
```
ratings.hist(column='rating',figsize=(10,5))
```

Out[45]:
```
array([[<Axes: title={'center': 'rating'}>]], dtype=object)
```



In [46]:
```
ratings.boxplot(column='rating',figsize=(10,5))
```

Out[46]: <Axes: >



In [47]:
```
tag.head()
```

Out[47]:

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	2009-04-24 18:19:40
1	65	208	dark hero	2013-05-10 01:41:18
2	65	353	dark hero	2013-05-10 01:41:19
3	65	521	noir thriller	2013-05-10 01:39:43
4	65	592	dark hero	2013-05-10 01:41:18

In [48]:
```
tag['tag'].head()
```

Out[48]:
```
0      Mark Waters
1        dark hero
2        dark hero
3    noir thriller
4        dark hero
Name: tag, dtype: object
```

In [49]:
```
ratings[-10:]
```

Out[49]:

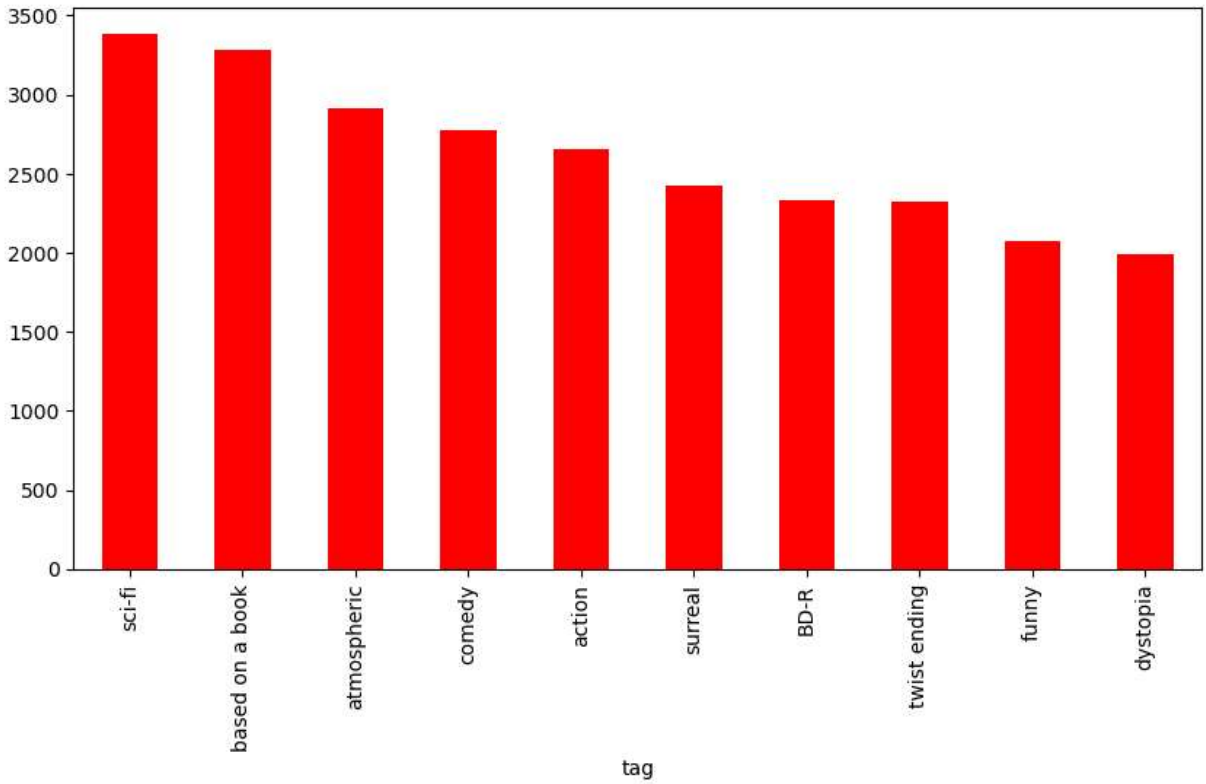
	userId	movieId	rating	timestamp
20000253	138493	60816	4.5	2009-12-03 18:32:43
20000254	138493	61160	4.0	2009-11-16 16:55:37
20000255	138493	65682	4.5	2009-10-17 21:52:53
20000256	138493	66762	4.5	2009-10-17 18:50:08
20000257	138493	68319	4.5	2009-12-07 18:15:20
20000258	138493	68954	4.5	2009-11-13 15:42:00
20000259	138493	69526	4.5	2009-12-03 18:31:48
20000260	138493	69644	3.0	2009-12-07 18:10:57
20000261	138493	70286	5.0	2009-11-13 15:42:24
20000262	138493	71619	2.5	2009-10-17 20:25:36

In [50]:
```
tag_counts=tag['tag'].value_counts()
tag_counts[-10:]
```

Out[50]:
```
tag
missing child                    1
Ron Moore                        1
Citizen Kane                     1
mullet                           1
biker gang                       1
Paul Adelstein                   1
the wig                          1
killer fish                      1
genetically modified monsters    1
topless scene                    1
Name: count, dtype: int64
```

In [52]:
```
tag_counts[:10].plot(kind ='bar',color= 'red',figsize=(10,5))
```

Out[52]:
```
<Axes: xlabel='tag'>
```



In [53]:
```
is_highly_rated=ratings['rating']>=5.0
ratings[is_highly_rated][30:50]
```

Out[53]:

	userId	movieId	rating	timestamp
239	3	50	5.0	1999-12-11 13:13:38
242	3	175	5.0	1999-12-11 13:32:13
244	3	223	5.0	1999-12-11 13:20:44
245	3	260	5.0	1999-12-11 13:09:02
246	3	316	5.0	1999-12-14 12:51:10
247	3	318	5.0	1999-12-11 13:09:26
248	3	329	5.0	1999-12-14 12:53:41
252	3	457	5.0	1999-12-11 13:16:55
253	3	480	5.0	1999-12-14 12:50:20
254	3	490	5.0	1999-12-11 13:30:41
256	3	541	5.0	1999-12-11 13:14:07
258	3	593	5.0	1999-12-11 13:16:55
263	3	858	5.0	1999-12-11 13:01:07
264	3	904	5.0	1999-12-11 13:04:10
267	3	924	5.0	1999-12-11 13:10:12
268	3	953	5.0	1999-12-11 13:11:52
271	3	1060	5.0	1999-12-11 13:18:30
272	3	1073	5.0	1999-12-11 13:20:44
275	3	1084	5.0	1999-12-11 13:15:03
276	3	1089	5.0	1999-12-11 13:05:58